

Amit Gohel

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Professional Summary

AI engineer with 2+ years shipping production LLM and agentic systems: permission-aware agentic RAG chatbot at 50,000+ queries/month (95% MRR), multi-model orchestration across GPT-4o, Gemini 2.5 Pro, and Qwen3-Coder with Pydantic-validated intermediate schemas, and on-premise document processing pipelines. Strong in Python, FastAPI, Milvus-backed retrieval, and agentic workflow design. Microsoft Azure AI certified.

Work Experience

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|---|-------------------------|---------------------|
| AI Engineer , Vivansh Infotech | <i>Ahmedabad, India</i> | Aug 2025 – Present |
| <ul style="list-style-type: none">Upgraded RAG chatbot to a fully agentic architecture using LangChain for a multi-tenant EHS platform (500+ concurrent users, 1,000+ queries/day) – integrated Milvus, PostgreSQL, and 10+ platform APIs for autonomous retrieval, workflow triggering, and multi-turn resolutionEnforced 250+ granular permissions per query via 2-stage retrieval (Milvus permission index + PostgreSQL cross-reference); GPT-4o handles access control and query rewriting across 10,000+ chunks; 80% MRRBuilt an AI-powered form builder that converts PDF forms to digital replicas in ~20 seconds (200+ forms generated in month 1), with bulk processing of 100+ PDFs in 5–10 minutes – replacing 1+ hour of manual work per formDesigned Pydantic-validated intermediate JSON schema enabling reliable LLM-to-platform conversion; multi-model pipeline (GPT-4o, Qwen3-Coder-480B, GPT-4o-mini) with each model selected via empirical benchmarking across blueprint generation tasksCreated an AI video course generation pipeline reducing production time from ~1 week to ~10 minutes (99% reduction) by orchestrating RAG-based video selection (Milvus + Cohere), Gemini 2.5 Pro for visual-audio alignment, ElevenLabs narration, and serverless FFmpeg rendering on AWS Lambda | | |
| AI/ML Developer , Silver Touch Technologies Ltd | <i>Ahmedabad, India</i> | Jun 2024 – Jul 2025 |
| <ul style="list-style-type: none">Rebuilt production RAG chatbot infrastructure serving 50,000+ queries/month across 10+ chatbot instances – migrated from in-code Llama-2-7b to VLLM-served API and replaced FAISS with Milvus, improving MRR by 111% (45% to 95%)Led vector database evaluation (Milvus vs. Qdrant vs. FAISS) on production-scale data; tested 6+ embedding models (BGE-M3, BGE-small, BGE-base, BGE-large, Stella, SPLADE) and built custom retrieval logic with BGE-M3 + rerankerReduced response latency 50% (16s→8s) via query preprocessing with intent classification and retrieval optimizationBuilt on-premise OCR pipeline for sensitive government documents (200+ page PDFs) using DocTR + DocLink merged via Llama-3.1-Nemotron-70b; ~90% entity extraction on Aadhaar, PAN, GST | | |
| Research Intern , Indian Institute of Technology (IIT) Delhi | <i>New Delhi, India</i> | Jan 2024 – May 2024 |
| <ul style="list-style-type: none">Built end-to-end air writing recognition system: ESP32 + MPU6050 wearable capturing 6-axis IMU data at 60Hz, classifying 62 alphanumeric characters via CNN-BiLSTM achieving 75% user-dependent accuracyCollected 12,400 labeled samples from 20 participants; systematically evaluated 1,000+ model configurations across 5 architecture families (Dense, CNN, LSTM, BiLSTM, hybrids) using TensorFlow/KerasApplied rigorous leave-one-subject-out cross-validation for user-independent evaluation; collaborated with researchers from IISc Bangalore and Howard University | | |

Technical Skills

- **Programming Languages:** Python, C++ , SQL
- **LLM and Agentic AI:** LangChain, GPT-4o, Gemini 2.5 Pro, Llama, Qwen3-Coder; RAG, Agentic Workflows, Prompt Engineering, Structured Output (Pydantic), Natural Language Processing (NLP)
- **Vector Databases and Retrieval:** Milvus, FAISS, Qdrant, BGE-M3, Cohere Embed/Rerank, SPLADE, OpenAI Embeddings
- **Infrastructure and DevOps:** FastAPI, VLLM, AWS (EC2, Lambda, S3, SAM), Docker, PostgreSQL, Git
- **AI/ML Frameworks:** PyTorch, TensorFlow/Keras, Hugging Face Transformers, Edge Impulse
- **Tools and APIs:** OpenAI API, Google Gemini API, Anthropic API, Together AI, ElevenLabs, FFmpeg, Selenium
- **Hardware/IoT:** Arduino, ESP32, NodeMCU, IMU Sensors, Raspberry Pi

Projects

Voicer – Sign Language to Speech Device (GitHub) 2023

- Developed wearable device translating Indian Sign Language to speech using 10 IMU sensors + NodeMCU with on-device TinyML inference in **2ms**, achieving **94.1% accuracy** across 12 gesture classes
- **Samsung Solve for Tomorrow – Top 10 among 70,000+ participants nationwide**; featured on CNN News18

WeepScope – Infant Cry Classifier (GitHub) 2023

- Built standalone device classifying infant cries into 5 categories using hybrid 1D CNN + GRU model on Arduino Nano 33 BLE Sense (1MB flash); achieved **86% test accuracy** with 5-cycle majority voting; research paper with Dr. Kirankumar Trivedi (Northeastern University)

Education

Bachelor of Engineering – Computer Engineering Jun 2020 – Jun 2024

Vishwakarma Government Engineering College, Ahmedabad, India

Certifications and Achievements

- **Microsoft Azure AI Engineer Associate** – Certified in designing and implementing Azure AI solutions
- **Samsung Solve for Tomorrow** – Top 10 finalist among 70,000+ participants nationwide; Voicer, wearable ISL-to-speech AI device; featured on CNN News18
- **Smart India Hackathon 2023** – Top 5 among 44,000+ teams nationwide (Hardware Edition, MIT Karnataka)
- **Competitive Programming:** 370+ problems solved in Python